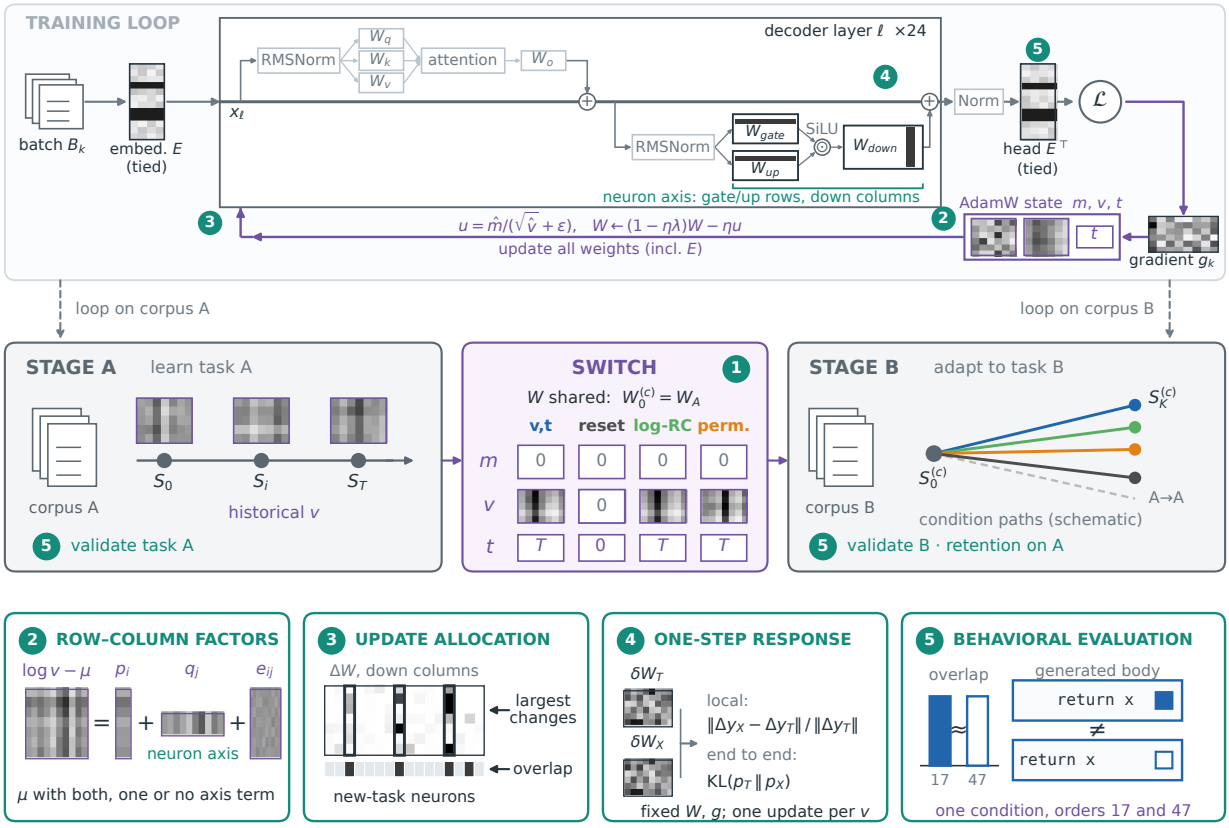

TRAINING LOOP
batch $B_k$
embed. $E$ (tied)
decoder layer $\ell$ ×24
RMSNorm
$W_q$
$W_k$
$W_v$
attention
$W_o$
$x_\ell$
RMSNorm
$W_{gate}$
$W_{up}$
SiLU
$W_{down}$
neuron axis: gate/up rows, down columns
Norm
head $E^\top$ (tied)
$\mathcal{L}$
AdamW state $m$, $v$, $t$
gradient $g_k$
$u = \hat{m}/(\sqrt{\hat{v}} + \varepsilon)$, $W \leftarrow (1 - \eta\lambda)W - \eta u$
update all weights (incl. $E$)
loop on corpus A
loop on corpus B
STAGE A
learn task A
corpus A
$S_0$
$S_i$
$S_T$
historical $v$
validate task A
SWITCH
$W$ shared: $W_0^{(c)} = W_A$
v,t
reset
log-RC
perm.
$m$
$v$
$t$
STAGE B
adapt to task B
corpus B
$S_K^{(c)}$
$S_0^{(c)}$
A→A
condition paths (schematic)
validate B · retention on A
ROW–COLUMN FACTORS
$\log v - \mu$
$p_i$
$q_j$
$e_{ij}$
neuron axis
$\mu$ with both, one or no axis term
UPDATE ALLOCATION
$\Delta W$, down columns
largest changes
overlap
new-task neurons
ONE-STEP RESPONSE
$\delta W_T$
$\delta W_X$
local:
$\|\Delta y_X - \Delta y_T\| / \|\Delta y_T\|$
end to end:
$\mathrm{KL}(p_T \| p_X)$
fixed $W$, $g$; one update per $v$
BEHAVIORAL EVALUATION
overlap
generated body
return x
≠
return x
17
47
one condition, orders 17 and 47